



BTEC Assignment Brief

Qualification	
Unit number and title	Unit 19: Business Intelligence
Learning aim(s) (For NQF only)	LO1 Discuss business processes and the mechanisms used to support business decision-making LO2 Compare the tools and technologies associated with business intelligence functionality LO3 Demonstrate the use of business intelligence tools and technologies LO4 Discuss the impact of business intelligence tools and technologies for effective decision-making purposes and the legal/regulatory context in which they are used.
Assignment title	Business Intelligence Dashboard for Adventure Works Cycles
Assessor	
Issue date	
Hand in deadline	

Vocational Scenario or Context	Project Context Adventure Works Cycles is a multinational bicycle manufacturer and retailer with offices in Canada, the UK, France, Australia and the United States. It sells road bikes, mountain bikes, touring bikes and accessories through Adventure Works stores (four locations), partner retailers. You were appointed as a junior business intelligence analyst by the company's Chief Commercial Officer three years after it began operations, at which point 2020-2022 sales data became available. He is weary of making business decisions solely from spreadsheet exports. He wants a Power BI dashboard to give executives and sales managers an up-to-date, single version of truth for Adventure Works business performance. He also wants you to write a professional report to explain and evaluate your Power BI design and implementation. Key stakeholders and their information needs: Chief Commercial Officer: Total revenue, profitability and performance against previous years.
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	Regional Sales Managers: Performance broken down by territory, country and continent. Product Managers: Sales figures by product category, category name, product return rate and product cost and revenue. Customer Insights Team: Analysis of demographic data and customer segmentation and behavior analysis. Provided Dataset: Adventure Works (2020-2022) Sales Data (fact table) - Contains OrderDate, CustomerKey, ProductKey, TerritoryKey, OrderQuantity and TotalRevenue. Returns Data - Contains ReturnDate, ProductKey, TerritoryKey and ReturnQuantity. Customer Lookup - Contains demographic data for each customer including Gender, Occupation, Education Level and Annual Income. Product Lookup + Categories + Subcategories - Contains the product name, ProductCost, ProductPrice and the CategoryName for each product. Territory Lookup - Contains the region, country and continent for each territory. Calendar Lookup - a full date dimension for time-intelligence calculations.
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Submission Requirements	Power BI Dashboard File .pbix Written Report .docx or .pdf
Checklist of evidence required	Task 1 – Business Context and BI Strategy (written report) 1. What is Business Intelligence and its role within retail businesses today? We are going to look at Adventure Works as a reference point throughout your discussion. 2. Describe and identify the main business processes in Adventure Works and diagram how the data is pushed through each process

(e. g. , sales, order management, returns, acquisition, customer experience, inventory).

3. Describe the five most important business questions you can answer using your dashboard with your top 5 KPIs. Explain the stakeholder, the question being answered and how your KPIs are calculated.

4. What is the most appropriate BI tool for Adventure Works from the following: Power BI, Tableau and Qlik Sense (Google Data Studio excluded). Make recommendations from this selection, justifying why. Consider the scale of Adventure Works, its data structure and its stakeholders.

5. Describe the data categories of structured, unstructured and semi-structured and provide some examples of each using the Adventure Works dataset.

Task 2 – Data Modelling and Pipeline (written report)

1. Describe your Power BI data model. Take a screenshot and add labels to show how tables are linked, naming the relationships, identifying Fact and Dimension tables, as well as the keys and relationship cardinalities. Take a screenshot with an annotated diagram showing these attributes.

2. Describe any steps that you had to take to prepare and clean data in Power BI or Power Query. Provide valid reasons as to why these were needed based on the accuracy, completeness, consistency rules.

3. Describe all your measures you created for the report, explain your reasoning for using the measure and explain why you feel this measure should be put in place for your stakeholders. For each measure, include a brief explanation of how your measure is used on your dashboard. Please include a minimum of five measures, suggestions: Total revenue, total sales orders, total profits, profit margin %, YTD revenue, MOM Revenue Growth%, Return Rate%, Average OrderValue, Revenue per customer.

Task 3 – Build Power BI Dashboard

Build an interactive Power BI dashboard with a minimum of four pages.

Each page must be designed for a specific stakeholder. Minimum requirements per page:



	Page 1 – Executive Overview (audience: CCO)
	KPI cards: Total Revenue, Total Orders, Total Profit, Profit Margin %
	Monthly revenue trend line chart (2020–2022)
	YoY revenue comparison bar chart
	Slicers for Year and Product Category
	Page 2 – Regional Sales Performance (audience: Regional Sales Manager)
	Filled map or bubble map showing revenue by country
	Clustered bar chart: revenue by continent and region
	Table: top 10 territories by revenue with conditional formatting
	Drill-through from country to individual order detail
	Page 3 – Product Performance (audience: Product Manager)
	Revenue and orders by Category and Subcategory (treemap or stacked bar)
	Return Rate % by product (bar chart with target reference line at 5%)
	Top 10 products by revenue (sorted bar chart)
	Profit Margin % by subcategory with colour-coded bands (high/medium/low)
	Page 4 – Customer Insights (audience: Customer Insights Team)
	Revenue by Occupation and Education Level (stacked bar or matrix)
	Revenue by Gender and Marital Status (donut or pie chart)
	Top 20 customers by revenue

DATA SOURCE FOR DASHBOARD:

Download files [here](#)

Assessment and Grading Criteria

Task	Pass	Merit	Distinction
T1	BI and key business processes are described with basic examples. KPIs are listed with a brief explanation. Power BI is mentioned as a tool.	KPIs are clearly mapped to specific stakeholders with business justification. Power BI is compared to at least two alternatives with reasoned justification of the choice.	A comprehensive BI strategy is presented with critical evaluation of BI tool options at operational, tactical, and strategic levels, supported by industry references.
T2	Star schema is described and screenshot included. At least 5 DAX measures are listed with their formulas. Basic Power Query steps are described.	Star schema is compared to normalised model with trade-off analysis. DAX measures are explained with business rationale. Data quality transformations are justified against data quality principles.	Advanced DAX (e.g., CALCULATE, time intelligence) is used and critically justified. Data modelling decisions are evaluated against scalability and enterprise BI performance requirements.
T3	Dashboard has 4 pages with required visuals. Slicers and filters are present. DAX measures are used in visuals. Charts have titles and labels.	Dashboard is user-friendly with a coherent theme. Drill-through is implemented. Visuals are correctly chosen for the data type and clearly labelled. Cross-filtering works correctly.	Dashboard demonstrates clear design rationale tailored to each stakeholder. Advanced features (e.g., bookmarks, tooltips, custom visuals) enhance UX. Evidence of user-feedback iteration is present.



Sources of information to support you with this Assignment	Textbooks Kimball, R. and Ross, M. (2013). The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling. 3rd ed. Wiley. Marr, B. (2015). Big Data: Using Smart Big Data, Analytics and Metrics to Make Better Decisions and Improve Performance. John Wiley & Sons. Ferrari, A. and Russo, M. (2019). The Definitive Guide to DAX. Microsoft Press. Jeston, J. and Nelis, J. (2014). Business Process Management. 3rd ed. Routledge. Online Resources Microsoft Power BI Documentation – https://learn.microsoft.com/en-us/power-bi/ DAX Guide – https://dax.guide/ SQLBI – https://www.sqlbi.com/ (DAX and data modelling tutorials) Journal of Big Data – https://journalofbigdata.springeropen.com/ IEEE Xplore Digital Library – https://ieeexplore.ieee.org/ Legislation and Frameworks UK General Data Protection Regulation (UK GDPR) – https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/ •Data Protection Act 2018 – https://www.legislation.gov.uk/ukpga/2018/12/contents ICO Accountability Framework – https://ico.org.uk/for-organisations/accountability-framework
Other assessment materials attached to this Assignment Brief	• ACM Digital Library – Transactions on AI and Data Science ACM Digital Library • Journal of Big Data Journal of Big Data • IEEE Xplore Digital Library Available at: https://journalofbigdata.springeropen.com/ Wiley Online Library. (2025). Information Systems Journal. Available at: https://onlinelibrary.wiley.com/journal/13652575